

Ayoub ELMORTAJI

Engineering Student — Cybersecurity & Cloud Computing

Final Year Internship (PFE) available from January 2027

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Location: Mohammedia, Morocco



Education

▪ ENSAM Casablanca, Morocco	2022 – 2027
▫ Engineering Degree — Cybersecurity & Cloud Computing	2024 – 2027
▫ Integrated Preparatory Classes	2022 – 2024

Professional Experience

AI Red Teaming Intern Jun. – Jul. 2026

Wana Corporate (INWI) — Casablanca, Morocco | Cybersecurity Red Teaming for AI

- Designed an internal AI assistant based on **RAG** with role-based access control (**RBAC**), then conducted a **Black Box red teaming** campaign (Garak, PyRIT, DeepTeam, Promptfoo) including a **RAG Corpus Poisoning** scenario.
- Uncovered **+11 critical vulnerabilities**.
- Hardened security through **defense-in-depth (Llama Prompt Guard 2 + LiteLLM AI Gateway)**, reducing **ASR from 40 % to 20 %**, and delivered a **prioritized remediation report** actionable by engineering teams.

OT Cybersecurity Intern Jul. – Aug. 2025

JESA S.A. — Casablanca, Morocco | Cyber-deception architecture for OT/SCADA systems

- Deployed a **cyber-deception** architecture (**Purdue model**, **PLC/SCADA honeypots**) coupled with **centralized monitoring (Wazuh, Suricata, OpenCTI)**, improving detection and analysis of industrial threats.
- Conducted **OT/SCADA penetration tests** and assessed vulnerabilities against **NIST SP 800-82** and **IEC 62443**, providing the company with a concrete assessment of its security posture.

Technical Projects

Private Cloud Platform with GitOps & DevSecOps

- Secured an end-to-end CI/CD pipeline on **k3s** by deploying a **private cloud platform (OpenStack, Terraform, Ansible, ArgoCD, Falco, Vault)** with runtime threat detection and automated security gates on every commit.

AI-Powered Threat Intelligence Platform — AWS

- Reduced **SOC analysis time from 4h to 5min** by designing a **serverless AWS** platform (Lambda, DynamoDB, Bedrock, VPC) collecting **IOCs/day**, with automated enrichment (**VirusTotal, MITRE ATT&CK**) and AI analysis (**Nova Lite**).

AWS FinOps AI Multi-Agent System — CrewAI

- Automated AWS cost optimization analysis by orchestrating **7 AI agents (CrewAI + Gemini 2.0 Flash)** across a full FinOps pipeline (Cost Explorer, PDF report, **REST API**, **S3 dashboard**).

Machine Learning-Based Intrusion Detection System

- Achieved **98 % detection accuracy** by developing an **ML-based IDS** (Logistic Regression, Random Forest, **XGBoost**), with **false positive reduction via PCA** and visualization on a **Streamlit dashboard**.

SOC Anomaly Detection — Wazuh & Machine Learning

- Improved **anomaly detection** by simulating attacks (**Atomic Red Team**), collecting logs via **Wazuh**, and developing a **Python risk scoring model** displayed on a **Streamlit dashboard**.

Malware Analysis Lab — Static & Dynamic Analysis

- Extracted malware behavior and **IOCs** by building an **isolated lab (FlareVM, REMnux, INetSim)** and conducting **static and dynamic analyses** documented in detailed reports.

Check Point Firewall Deployment — ENSAM Datacenter

- Strengthened access control and network segmentation by deploying **Check Point R82** in **high availability (ClusterXL)** on **EVE-NG / Proxmox** (Firewall, IPS, **IPsec/SSL VPN**, **Active Directory** integration).

SOC Platform — Network Supervision & Attack Simulation

- Validated detection rules across **15 Red/Blue Team scenarios** by deploying a **SOC platform (VMware, pfSense, Active Directory, Suricata, Wazuh)** for centralized supervision.

Technical Skills

- **AI Security / AI Red Teaming:** AI Pentest, LLM Red Teaming, RAG Security, Garak, PyRIT (Crescendo), DeepTeam, Promptfoo, Llama Prompt Guard 2, LiteLLM AI Gateway, Guardrails AI, AI Threat Modeling.
- **AI & Agents:** Ollama, Open WebUI, Amazon Bedrock, CrewAI, n8n, MCP, LLM, RAG.
- **Offensive Security:** OWASP Top 10, Burp Suite, OWASP ZAP, OT/SCADA Pentesting, Malware Analysis (FlareVM, REMnux).
- **Defensive Security & SOC:** SIEM (Wazuh, ELK, Splunk), IDS/IPS (Suricata, Snort), Threat Intelligence (OpenCTI), Atomic Red Team, Windows/Linux log analysis.
- **Cloud & Serverless:** AWS (EC2, Lambda, S3, IAM, Bedrock, DynamoDB, VPC, Secrets Manager, GuardDuty, CloudTrail, CloudWatch), Oracle Cloud (OCI).
- **Network & Network Security:** FortiGate, pfSense, OPNsense, VPN (IPsec/SSL), SD-WAN, EVE-NG.
- **Systems, Virtualization & DevSecOps:** Linux, Windows Server, Active Directory, Docker, Proxmox, OpenStack, Kubernetes (k3s), Terraform, Ansible, ArgoCD, Git, GitLab CI/CD.
- **Programming Languages:** Python, C.

Certifications

- **Cloud:** AWS Certified Cloud Practitioner, AWS Cloud Security Foundations, Oracle Cloud OCI Foundations Associate.
- **Security:** Fortinet NSE 5 (FortiSASE / SD-WAN 7.6), Blue Team Junior Analyst (BTJA), TryHackMe SOC Level 1.
- **AI:** TryHackMe AI Security, Agentic AI (DeepLearning.AI), Machine Learning Specialization (Coursera).

Awards & Distinctions

- **2nd place SecureIT CTF 2025** — Cisco Networking Academy Africa & Middle East ▪ **Top 1% TryHackMe.**
- **Excellence Award 2025 & 2026 (twice)** — Forum de l'Excellence, Hassan II University.
- **2nd place MCSC V13 CTF 2026 ENSIAS** (Beginner Track) ▪ **Semi-Finals MEA & CIS NetAcad Cup 2026.**

Soft Skills

- Project management ▪ Technical communication & knowledge sharing ▪ Analytical thinking & problem solving ▪ Self-driven ▪ Cross-functional teamwork ▪ Methodological rigor.

Languages

- Arabic (native) | French (professional) | English (professional).

Extracurricular Activities

- **The Purple HAT — Cybersecurity Club:** Project Manager 2024 – 2026
- **Sawaid AI Amal — Social Club:** Secretary General 2024 – 2026